

Due on monday, October 3rd, at the beginning of tutorial.

The questions indicated as [**SELF**] are for self-study and additional exercise:
do not submit your work on these questions.

----- ★ -----

1. Consider a two-state system, so that the Hilbert space is two-dimensional, i.e., the Hamiltonian is a 2×2 matrix and the wavefunction is a vector with two complex components. Let's denote the wavefunction as

$$\psi(t) = \begin{pmatrix} u(t) \\ v(t) \end{pmatrix}$$

We will consider a relatively simple (diagonal) Hamiltonian:

$$H = \begin{pmatrix} \epsilon_1 & 0 \\ 0 & \epsilon_2 \end{pmatrix}$$

- (a) [**6 pts.**] Using the Schroedinger equation, write down differential equations for the functions $u(t)$ and $v(t)$.
- (b) [**6 pts.**] Solve the differential equations, so that $u(t)$ and $v(t)$ are expressed in terms of the initial values $u(0)$ and $v(0)$.
- (c) [**5 pts.**] Show that the norm of the wavefunction, $|u(t)|^2 + |v(t)|^2$, is independent of time.
- (d) [**4 pts.**] Provide an expression for $\psi(t)$, in the case that the initial state is

$$\psi(0) = \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix}$$

2. Electromagnetic waves as particles:

- (a) Visible light has a wavelength in the range of 400-700 nm.
 - i. [**3 pts.**] What is the frequency range of visible light?
 - ii. [**5 pts.**] What is the energy range of a photon of visible light?
- (b) [**5 pts.**] A kitchen microwave operates at frequency 2.5 GHz and at power 750 Watts. How many photons does the microwave emit per second?

- (c) [**SELF**] How many microwave photons [from the previous question] are required to heat up a 0.2-liter glass of water by 10° ? The heat capacity of water is roughly $4185 \text{ J}/(\text{kg}\cdot\text{K})$, and a liter of water weighs roughly 1kg.
- (d) [**SELF**] A low-power laser (10^{-3} Watts) gives out light of wavelength 633 nm. How many photons does it emit per second?

3. Consider the Schroedinger equation for a single particle in one dimension:

$$i\hbar \frac{\partial}{\partial t} \psi(x, t) = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \psi(x, t) + V(x) \psi(x, t)$$

The potential $V(x)$ is real-valued, i.e., $V^*(x) = V(x)$. We will assume that the wavefunction is *normalized*.

- (a) [**10 pts.**] From the wavefunction one can obtain the probability density

$$\rho = |\psi(x, t)|^2 = \psi^*(x, t) \psi(x, t)$$

and the probability current density

$$j = \frac{\hbar}{2mi} \left[\psi^* \frac{\partial \psi}{\partial x} - \psi \frac{\partial \psi^*}{\partial x} \right] \quad \text{omitting arguments } (x, t).$$

Here ψ^* denotes the complex conjugate of ψ .

Using the Schroedinger equation, show that ρ and j obey the one-dimensional continuity equation

$$\frac{\partial j}{\partial x} = -\frac{\partial \rho}{\partial t}$$

- (b) [**3 pts.**] Write the equation describing what it means for the wavefunction to be ‘normalized’. (In case normalization has not been covered in class: look it up!) Write the equation in terms of $\psi(x, t)$ and also in terms of $\rho(x, t)$.
- (c) [**3 pts.**] Guess the form of the continuity equation for the probability density, for the case of a particle in three spatial dimensions.
- (d) [**SELF**] What does the continuity equation imply physically? The wikipedia page on “Continuity Equation” has a section called “quantum mechanics”. You might try reading through.